

Project Cover Page — AY 2026-2027

Katz School Faculty Research Initiative

Project Name

Embodied Compliance Council: Multi-Voice Deliberation, Calibration Discipline, and Spatial Audit for Regulated AI Workflows

Principal Investigators

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Funds Requested

\$3,000

(Single-year award targeting mean tier. Multi-year extension intended if Year 1 produces ICAIF 2027 acceptance and IEEE banking-paper submission per the project timeline.)

Project Summary (150 words max)

The Embodied Compliance Council is a research initiative producing a publicly-available, open-source AI deliberation infrastructure for regulated workflows. The system convenes five disagreement-by-design AI voices around each consequential decision (trading book changes in finance, loan and wealth-advisory decisions in banking), records each voice's verdict with confidence, and runs continuous Brier-score calibration to learn which voices' verdicts are load-bearing over time. A spatial extended-reality layer renders the council as a physical podium of voices, with cryptographic hash-chain audit of every adjudication. The project produces (1) a peer-reviewed publication at ICAIF 2027 on calibration discipline as audit primitive, (2) an IEEE banking paper on ECOA / Reg B disparate-impact detection via multi-voice review, and (3) an external NSF grant application building on Year-1 results. Two Katz graduate students participate as research assistants on banking risk math and spatial frontend implementation.

All relevant Department Chair / Program Director approvals:

[Signature block — Dr. NGO's department chair signs]

[Signature block — Prof. Yang's department chair signs]

Relevant Dean's or Deans' approval:

[Signature block — Katz Dean signs]